

Resume Analyzer Using Natural Language Processing (NLP)

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Abstract - Finding the best candidates for a position and informing users of their resume score and areas for growth are the two main objectives of Resume Screening. After a thorough analysis of the literature on current methods, it was found that while traditional systems like manual screening may lead to incorrect assumptions and the wastage of human potential, they are not robust in terms of processing, accuracy, or efficiency. Software must score the resumes of the candidates in real-time and match and rate them using machine learning and natural language processing techniques in order to obtain accuracy. Applications' resumes would be the input, and suggestions from the user side and an admin-side rated candidate resume list would be the output. By utilizing natural language processing techniques, output results are obtained instantly in real time. TF-IDF, Cosine Similarity NLP techniques were employed by the authors to match strings in the suggested system. It could be used in administrative agencies, government agencies, and multinational corporations where a large number of resumes need to be checked every day for multiple positions.

Keywords: Resume screening, Natural language processing (NLP), machine learning, Text classification

I. INTRODUCTION

The Resume Screening is an innovative software tool designed to streamline and enhance the resume screening process for both job seekers and employers. Exploiting latest natural language processing (NLP) and machine learning process, this application offers several key features and benefits. In current competitive interest market, corporations are inundated with a deluge of resumes for every interest posting. Resume screening, is a crucial initial step in the hiring process, can be time-consuming and prone to bias when done manually. To address these challenges, the Resume Screening App has emerged as a cutting-edge solution.

Resume screening will focus majorly on the design which will be used to find best candidate for a specific job post. The system will embolden the job applicant candidates as well as the recruiters to use it for job applications and find best candidate of resumes. Recruitment is a hectic procedure where the first task for any recruiter is to search the resumes.

Resume is then forwarded to the resume parser which is a pipeline of NLP techniques that will extract useful information. The job applicant as well as recruiters can make use of it for ease in applying for job openings and searching job applicant.

The recruiters from numerous groups can put up the information of the task openings to be hire of their respective groups. The resumes submitted by the candidates are matched with the job profile requirement posted by the company's recruiter by utilizing techniques like machine learning and Natural Language Processing (NLP).

Consequently, choosing qualified candidates from a field of numerous of candidate frequently is a challenging work for a corporate. In order to choose the best application from the pool, recruiters must sift through a lot of data. As a result, it greatly increases the burden of the relevant Recruiter department [5]. Furthermore, selecting the best candidate for a subsequent hiring procedure necessitates the use of significant human resources as well as intense work and resources. Recruiters may save a great deal of time and money if they can identify the non-relevant profiles early in the hiring process [10].

The benchmark flow will be in the following way, the applicant capitulates his Curriculum Vitae at the front- end the rise in software engineering jobs, with professional developers worldwide estimated to reach 28.7 million by 2024. The paper concludes that automatic software engineering position resume screening can significantly reduce the time, effort, and cost of hiring [1].

In order to categorize resumes according to job categories, automated resume classification can be used based on machine learning-based. This method can automate the boring process of selecting appropriate resume and support HR of company to overcome the above listed difficulties.

The techniques employed by these websites have produced results with good degrees of precision and accuracy; nevertheless, one disadvantage is the time needed to locate suitable applicants for a particular job description. Some of them have been effective in grouping candidate resumes for each job posting that each firm provides into distinct categories.

II. RELATED WORK

Resume screening today, in the digital age, is significantly more automated and data-driven compared to the older generation. Advanced software and AI tools are used to quickly process large volumes of resumes, filtering candidates based on specific keywords, skills, and qualifications [2]. This has led to a more efficient but also potentially biased process,

as algorithms may inadvertently favor certain demographics or overlook valuable candidates. Additionally, online platforms and professional networking sites have become crucial for jobseekers, allowing for broader reach but also intensifying competition. Overall, modern resume screening is characterized by speed, efficiency, and potential bias concerns, in contrast to the more manual and personal approach of the past.

In the past, recruiters manually reviewed resumes. However, with the increasing competition for roles, manual screening has become impractical.

Manual screening can lead to inefficiencies, with recruiters potentially missing out on suitable candidates or making mistakes in shortlisting due to various reasons. ATS automates the screening process, ensuring that all submitted resumes are evaluated and the best matches for the job description are selected.

ATS provides accuracy in shortlisting, ensuring that the most suitable candidates are selected based on their educational certificates and skills. The current systems are designed for companies, leaving applicants at a disadvantage. Even candidates with relevant skills may miss out on opportunities if they cannot present their qualifications effectively [7].

In relation to the suggested methodology, [11] may have the most relevant work. For this work, they implemented various classifier methods and compared the accuracy of each algorithm. The results were as follows—The 10-fold cross-validation accuracy for the RF classifier was 38.99%. We utilized another well-known classifier for this assignment, called "NB," because the results we got were not good enough. Compared to the previous classifier's (RF) accuracy, the NB classifier predicted the resume categories with a 44.39% prediction accuracy. But more than half of the resumes were incorrectly identified, according to the NB classifier's 44.39% accuracy rate. On the identical set of data, we employed a different classifier called "Linear SVM" and obtained an accuracy of 78.53%. A lower accuracy of 62.40% was obtained with the LR classifier—which was utilized to increase the model accuracy—than with the "Linear SVM" classifier.

The task of recruiting suitable person has become increasingly rigorous and killing time for corporates due to the growth of E-recruitment and candidates [8].

The accomplishment of talent is a key factor in ensuring business success, and manually filtering a classifying resumes submitted by a numerous number of candidates and matching them with the required job skills is a challenging task.

Known as text analytics, involves computational tools that transform and analyze unstructured text in documents using natural language processing (NLP) and machine learning algorithms.

These algorithms can perform tasks such as categorization, text tasks such as text categorization, classification, summarization, sentiment analysis, entity extraction, and concept identification, making text mining an essential technology in developing effective job-resume matching tools.

According to the experimental findings, a voting-based classifier generated 91.2% accuracy in forecasting the

categories, but in the absence of retraining, the accuracy was 84.2% [10].

This paper discusses the use of natural language processing for automatic software engineering position resume screening. The researchers focus on extracting technical skills using resume label character positioning, dataset consisting of candidate requirements, word and phrase matching for candidate information retrieval. Character positioning is introduced as a new technique for information extraction, which perceives needed data and pulls it out. This methodology creates a summary of the resume from the extracted information and computes count scores from recognized skills, education, and experience level. After testing on five random software engineering positions, the correct extraction rate of 33.59% was obtained. The paper highlights the importance of resume screening in the job market, particularly for candidate specialization in Information Technology and Software Engineering knowledge, where a large number of applicants apply worldwide.

Automatic resume screening uses several methods to effective by use of automated system for the analyzing of proper candidate for particular job. To offer a greater goal approach to lessen the opportunity of partiality in guide screening by utilizing cutting-edge NLP algorithms such as cosine similarity and S-BERT to improve resume screening accuracy.

III. METHODOLOGY

Assenting Fig.1, the total process is split into five parts

- (i) Data collection and visualization;
- (ii) Preprocessing:- Data Cleaning
- (iii) feature engineering; Feature Extraction
- (iv) building the model; Classification Technique
- (v) utilizing a graphical user interface (GUI) for model evaluation and testing in a real-time setting [9].

A. Preprocessing

The CVs that are being submitted as input will be cleaned during this procedure to get rid of any unusual or unnecessary characters. Special characters, digits, and single-letter words are all eliminated during cleaning. After completing these procedures, we were left with a clean dataset devoid of single letters or special characters. The NLTK tokenizes are used to divide the dataset into the tokens. Additionally, preprocessing techniques including stop word removal, stemming, and lemmatization are used to the tokenized dataset. After importing the raw CV file, the data in the resume field was cleaned to get rid of the superfluous spaces and numbers in the date.

Removing stop words: Words like and, the, was, and so on are often used in the text and are not useful for making predictions, thus they are eliminated. Procedure for removing stop words:

1. The input words have been tokenized into distinct tokens and are kept in an array.
2. At this point, every phrase corresponds to the list of stop words in the NLTK library:

/Imported Stop Word module from NLTK corpus/

- (a) From nltk.corpus import stop words
- (b) StopWords[] =

set(stopwords.words('english'))

- (c) It returns a total of 179 stopwords, which the print (StopWords) function can view and (len(StopWords)) can be used to verify.

3. If the words sieve from the important sentence array are comprise in the StopWords [] list.
4. Up until the tokenized array's final element is unmatched, the same procedure is repeated.
5. The resulting array contains no stopwords.

Stemming: The process of reducing word inflection to its basic forms, or stemming, involves assigning a set of words to a single stem even while the stem is not a recognized term in the language. The portion of a word that has inflectional (changing/deriving) affixes like (-ed, -ize, -s, -de, -ing, mis) is called the stem (root). The terms "playing," "plays," and "played," for instance, are mapped to their rootword "play," and the words "python," "pythoner," "pythoning," and "pythoned" are transferred to their base word "python."

Lemmatization: In contrast to stemming, lemmatization reduces the number of inflected phrases in order to guarantee that the root word accurately belongs in the language. Lemmatization consists of the following standard procedures:

- Create a wordlist from that extcorpus.
- Compile all of the word list items as they appear in the corpus to create a concordance of the corpus.
- Using the concordance, assign the word-forms to their corresponding lemmas.

B. Feature Extraction

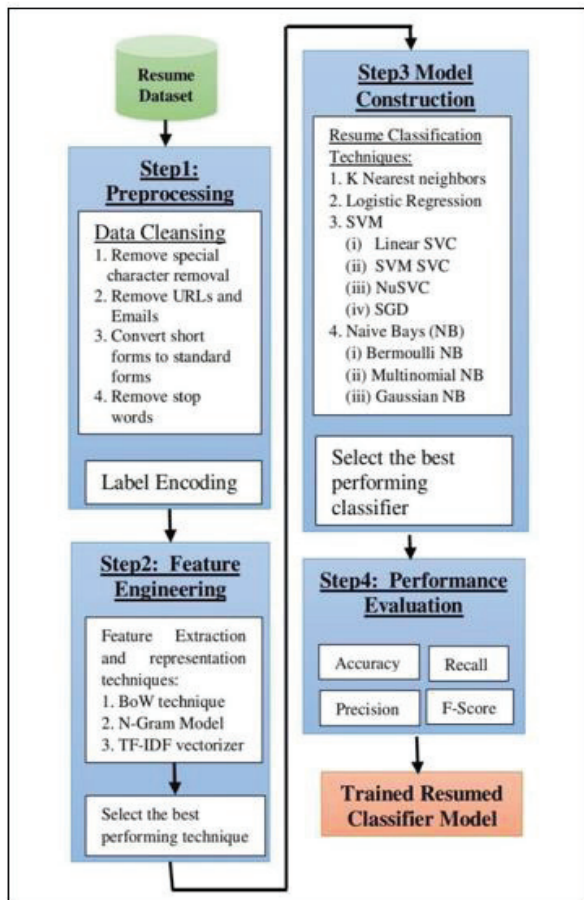


Fig. 1. Method for Resume Classification

Tf-Idf can be used to extract features from a preprocessed dataset. Tf-Idf was used to perform feature extraction on the cleaned data once it had been imported. To handle the input, the machine learning- based categorization model or learning algorithms require a numerical vector of a specified size. Text with varying lengths was not processed by ML-based classification tools. Because of this, the texts are converted into the appropriate vector form of similar length during the preparation phases. A few techniques for extracting the features are term frequency, inverse document frequency, and bag of words (BoW). The BoW model states that each document's narrative in this instance, the terms' existence (and often their frequency).

Model Construction

Each text classifier has a different philosophy for classifying the examples, nine different text classifiers were used. For multiclass classification, the "One-Vs-Rest-Classification" technique was applied.

The following is a synopsis of the nine machine learning models that have been put into practice:

- a) K Nearest Neighbors (KNN): KNN assigns a label based on the highest adjacent data point after determining the k-nearest data points to the new instance. Because of its simplest approach of using the Euclidian distance equation for classification problems, KNN is also known as a lazy learner classifier.
- b) Multinomial Naïve Bayes (MNB): The conditional probability serves as the foundation for the Naïve Bayes classifier. The likelihood that a vector belongs to a class is determined by the NB classifier. It determines the likelihood for each of the provided examples and uses the conditional probability to classify. Strong independence between the characteristics serves as its foundation. One Naïve Bayes variation that uses a multinomial distribution for all pairs is called MNB.
- c) Logistic Regression (LR): With a
- d) threshold value, Logistic Regression uses the logistic function to the classification task. For classification issues, LR is thought to be one of the simplest implementation [11].
- e) Support Vector Classifier (SVC): SVC uses the Kernel concept to overcome the drawbacks of Linear SVM, which is particularly useful for non-linearly separable data.
- f) Linear Support Vector Classifier (LSVC): This model relies on identifying the optimal line that divides two classes. The simplest type of support vector machine is this one, which locates the linear hyperplane that connects two classes. Nevertheless, if the data cannot be separated linearly, it will not produce meaningful findings. The least square Support Vector Machine classifier is another name for linear SVM.
- g) Nu-Support Vector Classifier (Nu SVC): This model is comparable to the SVC in that it regulates the number of support vectors using a parameter.

- h) Stochastic Gradient Descent (SGD): It looks for the minima of the loss using SGD, and it employs SGD for training.

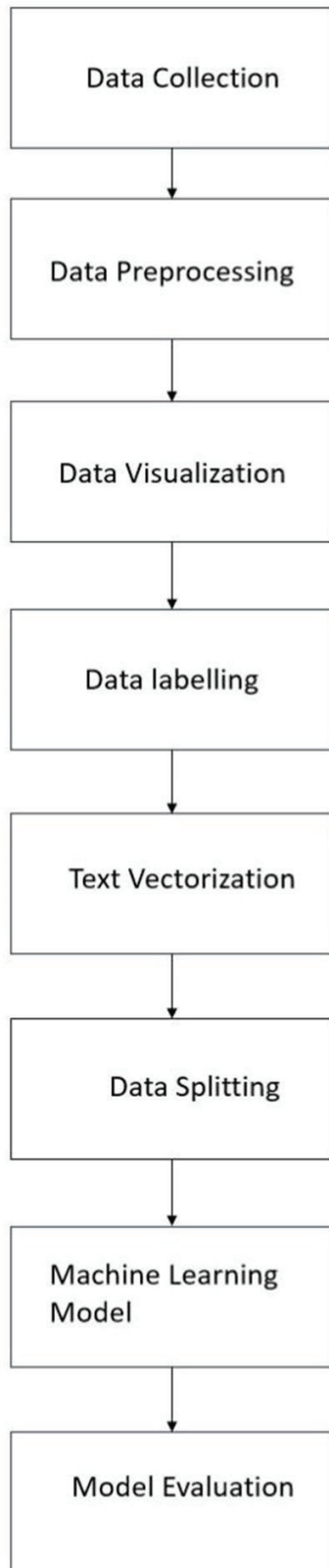


Fig. 2. Data Processing

For later analysis and testing, the collected features and taught machine learning models were saved in a Python external pkl file format. The extracted feature representation and learned models were stored on disk using the scikit-learn

externals module, and subsequently used in the GUI for real-time testing.

C. Performance Evaluation

We employed various performance calculation pattern to gauge the effectiveness of the a fore mentioned classification models. Because of the imbalance in the dataset, the inclusive definiteness was not the only important matrix for assessing the model.

The Evaluation step assesses the accuracy of the NLP algorithms used for resume analysis and considers the impact of formatting or consistency issues on information extraction [3].

Consequently, matrices for overall accuracy, precision, recall, and F-score were employed to evaluate performance.

1. Overall Accuracy: The percentage of predictions that the algorithms properly identify is known as accuracy. When dealing with imbalanced data, accuracy by itself does not provide a whole picture.
2. Precision: Precision looks for the percentage of positive forecasts that are actually true. Precision has a value between 0 and 1. A model has a precision of 1 if it does not yield false positive findings. It provides insight into the accuracy with which the model is able to pinpoint the True positive values of classes. Results in the multiclass classification problem have a precision of 1 for all classes.
3. Recall: This type of recall aims to determine how much proportion of the real positive records is accurately detected. Precision has a value between 0 and 1. A model with a precision of 1 does not generate false-negative outcomes. In a multiclass classification problem, the calculated. The average calculates the statistic for every class separately before averaging the results
4. F-Score: The weighted harmonic mean of the test's precision and recall determines the F- Measure. The values range from 0 to 1, with the greatest number (1), indicating that the algorithm has reached its optimal levels of precision and recall.

IV. RESULT

The user interface consists of user panel and admin panel. Interface is made using streamlit which is a python built-in library. Regular user side panel has the option to browse and upload the resume as the screen in Figure 3.

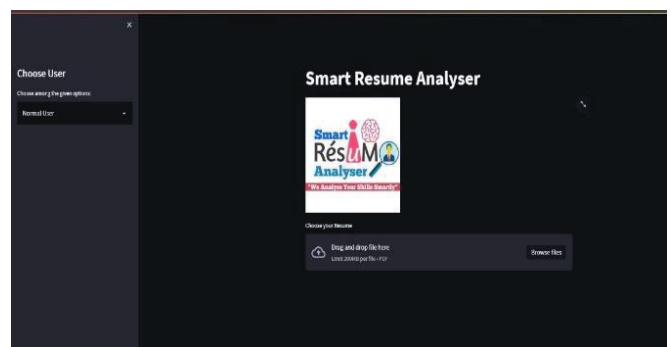


Fig. 3. User side interface

[illegible]

Admin side in Figure 5 consists of login form where the HR/ recruiter's authentication can be verified.



User's Data

	Name	Email	Row	Timestamp	Total	Predicted Field
0	Harsh Ghogale	ghogaleharsh46@gmail.com	20	2024-02-27_14-09:37	2	
1	art director	hello@allisonbeer.com	20	2024-02-27_14-14:33	1	Web Development
2	Harsh Ghogale	ghogaleharsh46@gmail.com	20	2024-02-27_14-15:41	2	
3	art director	hello@allisonbeer.com	20	2024-02-27_14-19:43	1	Web Development
4	art director	hello@allisonbeer.com	20	2024-02-27_14-19:55	1	Web Development
5	Android Developer	info@qwikresume.com	40	2024-02-27_14-25:00	2	Android Development
6	Android Developer	info@qwikresume.com	40	2024-02-27_14-25:24	2	Android Development
7	Android Developer	info@qwikresume.com	40	2024-02-27_14-25:35	2	Android Development
8	Data Scientist	info@qwikresume.com	40	2024-02-27_14-26:05	2	Data Science
9	Harsh Ghogale	ghogaleharsh46@gmail.com	20	2024-02-27_14-43:15	2	

Pie-Chart for Predicted Field Recommendations

Predicted Field according to the Skills

Field	Percentage
0	63.6%
Web Development	25.8%
Android Development	7.02%
Data Science	2.46%

The job description and the qualifications listed in the resume determine the user's experience level, which is represented in the pie chart below. Based on the job description, the administrator uses the experience level to focus on the kind of employee they are looking for.



1. This approach will continue to be developed by analyzing applicant social networking data (such as their GitHub, LinkedIn, and Facebook sites) and combining it with resume material to generate even more insightful recommendations.
2. We want to use the data collected from candidate resumes in our ongoing study to dynamically develop user profiles that will be used to recommend jobs to job seekers.
3. By using the social media information presented on their applications, the personalities of the shortlisted applicants can be investigated. This assessment will assist in determining whether the candidate's social life personality matches the requirements of the role.

The important topic of automatic resume quality assessment (RQA) was examined in this research. Classifying an applicant's resume is a tedious, resource-wasting, and time-consuming exercise. To solve this problem, automated system is created that uses machine learning to suggest resumes of eligible applicants to HR based on the job description that is given. Our system successfully screened and narrowed down the pool of most eligible applicants by utilizing Natural Language Processing techniques [2].

In future scope it can be make quality of system in such a way that the system does most of the scrutinizing procedure of resumes. We can elicit the expertise and experience and works of the candidate from the Linked in profiles mentioned in the resume. Hence, it makes the recruitment process easier and smoother. We can further upgrade it to system so that it

can extract information from the resumes when uploading in different formats.

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